

**HANKO TULLINIEMI, Finland**WMO: **027460**Lat: **59.809N**Lon: **22.916E**Elev: **28**StdP: **101.00**Time Zone: **2.00 (EUE)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |             |            | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|-------------|------------|-----------------------|-----------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%          |            |                       |           |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS          | MCDB       | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-15.7 | (c)<br>-13.0 | (d)<br>-17.9                  | (e)<br>0.8 | (f)<br>-15.3 | (g)<br>-15.3 | (h)<br>1.0 | (i)<br>-12.1 | (j)<br>20.1           | (k)<br>-0.1 | (l)<br>18.6 | (m)<br>0.7 | (n)<br>6.2            | (o)<br>50 | (p)<br>0.873 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD<br>to 0.4% DB |           |
|---------------|---------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------------|-----------|
|               |                           | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                         |           |
|               |                           | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                    | PCWD      |
| (a)<br>7      | (b)<br>4.1                | (c)<br>24.0     | (d)<br>20.5 | (e)<br>22.5 | (f)<br>19.4 | (g)<br>21.1 | (h)<br>18.2 | (i)<br>20.9         | (j)<br>23.3 | (k)<br>19.8 | (l)<br>22.1 | (m)<br>18.6 | (n)<br>20.8 | (o)<br>6.7              | (p)<br>80 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.9                            | 14.7 | 22.7 | 18.9 | 13.7 | 21.6 | 17.6 | 12.6 | 20.2 | 60.6          | 23.3 | 56.9 | 22.2 | 52.9 | 20.8 | 23.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 17.7              | 15.9 | 14.4 | -16.4                      | 25.0 | 3.9                | 2.6 | -19.2                                              | 26.9 | -21.5      | 28.5 | -23.7      | 29.9 | -26.6      | 31.8 |
|                   |      |      | -16.7                      | 21.1 | 3.8                | 1.9 | -19.5                                              | 22.5 | -21.7      | 23.6 | -23.9      | 24.7 | -26.7      | 26.1 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

**Monthly Climatic Design Conditions**

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 7.0   | -2.8  | -2.6  | -0.5  | 3.6   | 9.5   | 13.9  | 18.0  | 17.5  | 13.8  | 7.9   | 4.4   | 0.9  | (6)  |
| (7)  |                                                                                  | DBStd       | 7.99  | 4.53  | 4.98  | 3.39  | 2.45  | 3.47  | 2.34  | 2.81  | 2.09  | 2.54  | 3.06  | 3.61  | 4.30 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1850  | 396   | 353   | 325   | 191   | 52    | 1     | 0     | 0     | 4     | 77    | 169   | 282  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 4175  | 654   | 586   | 583   | 441   | 272   | 135   | 42    | 41    | 137   | 324   | 419   | 540  | (9)  |
| (10) |                                                                                  | CDD10.0     | 763   | 0     | 0     | 0     | 1     | 38    | 117   | 247   | 232   | 118   | 11    | 0     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 46    | 0     | 0     | 0     | 0     | 0     | 1     | 31    | 14    | 0     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 66    | 0     | 0     | 0     | 0     | 0     | 1     | 61    | 4     | 0     | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 7.6   | 7.9   | 7.8   | 7.9   | 7.0   | 6.9   | 6.9   | 6.4   | 6.6   | 7.5   | 8.2   | 8.8   | 9.6  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 657   | 61    | 43    | 40    | 35    | 29    | 41    | 57    | 75    | 55    | 76    | 80    | 66   | (15) |
| (16) |                                                                                  | PrecMax     | 762   | 108   | 98    | 93    | 63    | 62    | 77    | 166   | 176   | 145   | 177   | 163   | 120  | (16) |
| (17) |                                                                                  | PrecMin     | 478   | 8     | 3     | 5     | 9     | 5     | 14    | 1     | 7     | 15    | 30    | 7     | 19   | (17) |
| (18) |                                                                                  | PrecStd     | 81    | 27    | 26    | 19    | 15    | 16    | 18    | 43    | 41    | 33    | 35    | 39    | 31   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 5.0   | 4.3   | 6.4   | 12.7  | 19.5  | 21.1  | 26.7  | 24.2  | 19.3  | 14.2  | 10.4  | 8.1  | (19) |
| (20) |                                                                                  |             | MCWB  | 4.1   | 2.6   | 3.3   | 8.1   | 13.6  | 16.9  | 20.7  | 20.5  | 16.5  | 12.5  | 9.8   | 7.2  | (20) |
| (21) |                                                                                  | 2%          | DB    | 4.2   | 3.1   | 4.7   | 10.0  | 17.7  | 19.6  | 25.0  | 22.6  | 18.4  | 13.2  | 9.7   | 7.2  | (21) |
| (22) |                                                                                  |             | MCWB  | 3.5   | 2.2   | 2.7   | 6.7   | 13.0  | 15.5  | 20.4  | 19.6  | 16.0  | 11.9  | 9.1   | 6.3  | (22) |
| (23) |                                                                                  | 5%          | DB    | 3.4   | 2.5   | 3.9   | 8.3   | 16.2  | 18.4  | 23.7  | 21.5  | 17.4  | 12.5  | 9.0   | 6.5  | (23) |
| (24) |                                                                                  |             | MCWB  | 2.8   | 1.9   | 2.2   | 5.5   | 12.5  | 14.8  | 20.3  | 18.6  | 15.4  | 11.4  | 8.2   | 5.6  | (24) |
| (25) |                                                                                  | 10%         | DB    | 2.6   | 2.1   | 3.1   | 6.9   | 14.7  | 17.3  | 22.3  | 20.5  | 16.7  | 11.8  | 8.3   | 5.8  | (25) |
| (26) |                                                                                  |             | MCWB  | 2.0   | 1.6   | 2.0   | 4.7   | 11.7  | 14.3  | 19.3  | 17.7  | 14.9  | 10.7  | 7.4   | 4.8  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 4.5   | 3.1   | 3.9   | 8.4   | 14.9  | 17.9  | 22.2  | 21.2  | 17.5  | 13.1  | 10.0  | 7.6  | (27) |
| (28) |                                                                                  |             | MCDWB | 4.8   | 3.5   | 5.4   | 12.2  | 18.2  | 20.2  | 24.6  | 23.1  | 18.9  | 13.6  | 10.1  | 7.9  | (28) |
| (29) |                                                                                  | 2%          | WB    | 3.7   | 2.5   | 3.2   | 6.9   | 13.8  | 16.6  | 21.5  | 20.1  | 16.5  | 12.4  | 9.3   | 6.5  | (29) |
| (30) |                                                                                  |             | MCDWB | 4.1   | 2.9   | 4.2   | 9.5   | 16.6  | 18.9  | 24.0  | 22.1  | 17.6  | 12.9  | 9.6   | 7.0  | (30) |
| (31) |                                                                                  | 5%          | WB    | 2.9   | 2.1   | 2.7   | 5.9   | 12.8  | 15.6  | 20.7  | 19.3  | 15.8  | 11.7  | 8.5   | 5.7  | (31) |
| (32) |                                                                                  |             | MCDWB | 3.4   | 2.5   | 3.5   | 7.9   | 15.5  | 17.6  | 23.2  | 21.1  | 16.9  | 12.3  | 8.9   | 6.3  | (32) |
| (33) |                                                                                  | 10%         | WB    | 2.1   | 1.7   | 2.2   | 5.0   | 11.9  | 14.8  | 19.6  | 18.2  | 15.3  | 10.8  | 7.6   | 4.9  | (33) |
| (34) |                                                                                  |             | MCDWB | 2.5   | 2.0   | 2.9   | 6.5   | 14.3  | 16.7  | 22.1  | 20.0  | 16.4  | 11.6  | 8.1   | 5.7  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 3.9   | 3.3   | 3.9   | 3.9   | 4.4   | 4.0   | 4.1   | 3.5   | 3.1   | 3.0   | 2.6   | 3.1  | (35) |
| (36) |                                                                                  |             | MCDBR | 2.7   | 2.4   | 3.6   | 5.8   | 6.3   | 5.4   | 4.9   | 4.1   | 3.6   | 2.8   | 2.6   | 2.7  | (36) |
| (37) |                                                                                  |             | MCWBR | 2.8   | 2.2   | 2.6   | 3.5   | 3.8   | 3.1   | 3.0   | 2.7   | 2.5   | 2.7   | 2.8   | 2.9  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 2.6   | 2.4   | 3.2   | 5.2   | 6.2   | 4.7   | 4.6   | 3.4   | 3.0   | 2.6   | 2.5   | 2.7  | (38) |
| (39) |                                                                                  |             | MCWBR | 2.8   | 2.2   | 2.5   | 3.3   | 4.1   | 3.1   | 3.1   | 3.0   | 2.4   | 2.8   | 2.8   | 3.1  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.281 | 0.313 | 0.333 | 0.371 | 0.366 | 0.362 | 0.389 | 0.382 | 0.351 | 0.334 | 0.311 | 0.267 | (40) |      |
| (41) |                                                                                  | taud        | 2.168 | 2.235 | 2.300 | 2.325 | 2.422 | 2.476 | 2.421 | 2.432 | 2.498 | 2.407 | 2.218 | 2.099 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 490   | 681   | 795   | 828   | 862   | 872   | 834   | 811   | 780   | 661   | 446   | 350   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 44    | 71    | 90    | 105   | 103   | 99    | 103   | 94    | 75    | 60    | 42    | 31    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.24  | 0.77  | 2.29  | 3.96  | 5.55  | 5.86  | 5.53  | 4.36  | 2.65  | 1.15  | 0.31  | 0.15  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.04  | 0.14  | 0.27  | 0.41  | 0.54  | 0.38  | 0.47  | 0.43  | 0.32  | 0.16  | 0.06  | 0.02  | (45) |      |

**Historical Trends**

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (59 neighbors) |  | +0.39 | N/A     | N/A    | N/A     | N/A   | N/A   | -129        | N/A     | N/A     | N/A     |

Nomenclature: See separate page